



HEALTHCARE TITANS

Hackathon 2026

AI IN HEALTHCARE

AI Digital Wellbeing & Healthcare Assistant

An AI-powered platform for real-time health monitoring, intelligent diagnosis, and personalized wellbeing insights.

Team Healthcare Titans

TEAM LEADER

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INSTITUTION

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PROBLEM STATEMENT

The Hidden Crisis in Digital Wellbeing



No Proactive Health Monitoring

People only visit doctors when already ill. Lifestyle habits like poor sleep, high screen time, and low exercise go untracked and ignored.



Late & Expensive Medical Attention

Symptom identification requires clinical visits. Mental health issues like stress and burnout are detected far too late for early intervention.



No Personalized Guidance

Generic health advice does not account for individual lifestyle patterns. Users need an AI that understands their unique health context.

THE SCALE OF THE CRISIS

71%

of adults experience unhealthy stress levels with no early detection system

50%

of chronic diseases are preventable with early lifestyle intervention

6.5B

smartphone users who could benefit from AI-powered health monitoring

AI Digital Wellbeing & Healthcare Assistant

A full-stack AI platform built with Python Flask, SQLite, MediaPipe, and Chart.js — delivering real-time health insights to every user.



Health Score Engine

AI-calculated daily health score from 6 key metrics: sleep, screen time, mood, exercise, water intake, and work hours. Provides instant actionable feedback.

Python Flask + SQLite



AI Health Chatbot

Intelligent conversational AI for personalized health advice. Maintains full conversation history and responds to health queries 24/7 without clinical visits.

Flask + SQLite Chat History



AI Stress Detection

Real-time stress detection using device camera. Powered by MediaPipe AI Vision for facial landmark analysis — no wearables needed, runs entirely in-browser.

MediaPipe (Google AI Vision)



Symptom Checker

AI-powered symptom analysis that suggests possible conditions and next steps. Reduces unnecessary ER visits and helps users make informed decisions about seeking care.

Flask ML + HTML/CSS/JS



Analytics Dashboard

Visual health analytics with interactive Chart.js graphs tracking sleep, exercise, mood, and health score trends over 7/14/30 day periods.

Chart.js + Bootstrap 5



Secure Auth System

Full user authentication with bcrypt password hashing, session management, and personal health history — each user's data is private and persistent.

Werkzeug + SQLite

Who Benefits from This Platform?



Students & Young Adults

Track exam-period stress, irregular sleep, and poor nutrition. The AI chatbot guides students on managing academic pressure and maintaining healthy habits during high-workload periods.



Working Professionals

Monitor work-life balance through screen time, work hours, and physical activity tracking. The stress detection camera feature helps identify burnout before it becomes critical.



Elderly & Chronic Patients

Symptom checker provides guidance for age-related conditions. Regular health data logging with AI-backed recommendations helps manage long-term health without constant hospital visits.



Healthcare Institutions

Hospitals and clinics can deploy the platform for pre-appointment screening and post-discharge monitoring. Aggregate analytics help identify population health trends.

Zero

Hardware Required

24/7

AI Availability

Any

Device, Browser-based

Scalable & Open Source Ready

The Flask + SQLite architecture can be scaled to PostgreSQL or MySQL for enterprise deployment, making it accessible from rural clinics to urban hospitals.

Building the Future of Preventive Healthcare



Reduce Preventable Hospital Visits

Early AI detection of stress and lifestyle imbalances prevents escalation into clinical conditions, directly reducing the burden on India's healthcare system.



Democratize Healthcare Access

A browser-based platform with no hardware requirements makes AI health monitoring accessible to Tier 2/3 cities and rural populations across India.



Future: IoT & Wearable Integration

Planned expansion to integrate smartwatch data, IoT sensors, and real doctor consultation booking directly from the platform dashboard.



Multilingual & Offline Support

Roadmap includes Hindi and regional language support plus offline mode to serve patients in low-connectivity areas with AI-based areas with AI-based health guidance.

TECH STACK

Backend **Python Flask**

Database **SQLite (MySQL ready)**

AI Vision **MediaPipe (Google)**

Frontend **HTML/CSS/JS + Bootstrap 5**

Charts **Chart.js v4**

Auth **Werkzeug Bcrypt**

"Prevention is better than cure"

We built this platform to make AI-powered preventive healthcare accessible to every Indian — from students to seniors — using technology that runs on any smartphone or laptop.

— Team Healthcare Titans, MUIT NOIDA